

Assignment No: – 5

ON

Full Stack Development – SSCA4020

AIM: Perform CRUD operation using MongoDB and Server.

BACHLEOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)

SUBMITTED BY:

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Class: BSCIT-H – 7A - Batch 2023-27



CODE:

Commands

```
npm init -y
npm install mongodb mysql2
node mongo_crud.js
node mysql_crud.js
```

mongo_crud.js

```
const { MongoClient } = require("mongodb");

const uri = "mongodb://127.0.0.1:27017";
const client = new MongoClient(uri);

async function run() {
  try {
    await client.connect();
    console.log("Connected to MongoDB");
    const col = client.db("fsd_lab").collection("students");
    await col.deleteMany({});

    // ----- CREATE -----
    const ins = await col.insertMany([
      { name: "Rishabh Shah", enrollment: "23SS02IT181", sem: 7 },
      { name: "Aarav Patel", enrollment: "23SS02IT101", sem: 7 },
      { name: "Riya Mehta", enrollment: "23SS02IT145", sem: 7 },
    ]);
    console.log("\n[CREATE] Inserted documents:", ins.insertedCount);

    // ----- READ -----
    console.log("\n[READ] All students:");
    console.log(await col.find({}, { projection: { _id: 0 } }).toArray());

    // ----- UPDATE -----
    const upd = await col.updateOne(
      { enrollment: "23SS02IT101" },
      { $set: { sem: 8 } }
    );
    console.log("\n[UPDATE] Matched:", upd.matchedCount, "Modified:", upd.modifiedCount);
    console.log(await col.findOne({ enrollment: "23SS02IT101" }, { projection: { _id: 0 } }));

    // ----- DELETE -----
    const del = await col.deleteOne({ enrollment: "23SS02IT145" });
    console.log("\n[DELETE] Deleted:", del.deletedCount);

    console.log("\n[READ] After delete:");
    console.log(await col.find({}, { projection: { _id: 0 } }).toArray());
  } finally {
    await client.close();
  }
}
run().catch(console.error);
```

mysql_crud.js

```
const mysql = require("mysql2/promise");

async function run() {
  // Connect to MySQL server
  const conn = await mysql.createConnection({
    host: "localhost",
    user: "root",
    password: "root",
  });
  console.log("Connected to MySQL");

  await conn.query("CREATE DATABASE IF NOT EXISTS fsd_lab");
  await conn.query("USE fsd_lab");
  await conn.query("DROP TABLE IF EXISTS students");
  await conn.query("CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    enrollment VARCHAR(20) UNIQUE,
    sem INT
```

```

    `);
    console.log("Table 'students' created");

    // INSERT
    await conn.query(
      "INSERT INTO students (name, enrollment, sem) VALUES ?",
      [
        ["Rishabh Shah", "23SS02IT181", 7],
        ["Aarav Patel", "23SS02IT101", 7],
        ["Riya Mehta", "23SS02IT145", 7],
      ]
    );
    console.log("3 rows inserted");

    // SELECT
    let [rows] = await conn.query("SELECT * FROM students");
    console.log("\nSELECT * FROM students");
    console.table(rows);

    // UPDATE
    const [u] = await conn.query("UPDATE students SET sem = 8 WHERE enrollment = ?",
    ["23SS02IT101"]);
    console.log("\nUPDATE -> affected rows:", u.affectedRows);

    // DELETE
    const [d] = await conn.query("DELETE FROM students WHERE enrollment = ?", ["23SS02IT145"]);
    console.log("DELETE -> affected rows:", d.affectedRows);

    // SELECT with condition
    [rows] = await conn.query("SELECT name, sem FROM students WHERE sem >= ? ORDER BY name",
    [7]);
    console.log("\nSELECT name, sem FROM students WHERE sem >= 7 ORDER BY name");
    console.table(rows);

    await conn.end();
  }
  run().catch(console.error);

```

OUTPUT:

```

Windows PowerShell - MongoDB CRUD

PS C:\Users\Rishabh\FSD\practical7> npm install mongodb mysql2
added 24 packages, and audited 25 packages in 4s

found 0 vulnerabilities
PS C:\Users\Rishabh\FSD\practical7> node mongo_crud.js
Connected to MongoDB

[CREATE] Inserted documents: 3

[READ] All students:
[
  { name: 'Rishabh Shah', enrollment: '23SS02IT181', sem: 7 },
  { name: 'Aarav Patel', enrollment: '23SS02IT101', sem: 7 },
  { name: 'Riya Mehta', enrollment: '23SS02IT145', sem: 7 }
]

```

```

Windows PowerShell - MongoDB CRUD (contd.)

[UPDATE] Matched: 1 Modified: 1
{ name: 'Aarav Patel', enrollment: '23SS02IT101', sem: 8 }

[DELETE] Deleted: 1

[READ] After delete:
[
  { name: 'Rishabh Shah', enrollment: '23SS02IT181', sem: 7 },
  { name: 'Aarav Patel', enrollment: '23SS02IT101', sem: 8 }
]

```

```

Windows PowerShell - MySQL CRUD

PS C:\Users\Rishabh\FSD\practical7> node mysql_crud.js
Connected to MySQL
Table 'students' created
3 rows inserted

SELECT * FROM students

  (index) | id | name       | enrollment | sem |
  ---|---|---|---|---|
  0      | 1  | 'Rishabh Shah' | '23SS02IT181' | 7 |
  1      | 2  | 'Aarav Patel'  | '23SS02IT101' | 7 |
  2      | 3  | 'Riya Mehta'   | '23SS02IT145' | 7 |

UPDATE -> affected rows: 1
DELETE -> affected rows: 1

SELECT name, sem FROM students WHERE sem >= 7 ORDER BY name

  (index) | name       | sem |
  ---|---|---|
  0      | 'Aarav Patel' | 8 |
  1      | 'Rishabh Shah' | 7 |
  
```

```

MySQL Command Line Client

PS C:\Users\Rishabh\FSD\practical7> mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
mysql> USE fsd_lab;
Database changed
mysql> SELECT * FROM students;
+-----+-----+-----+-----+
| id | name       | enrollment | sem |
+-----+-----+-----+-----+
| 1 | Rishabh Shah | 23SS02IT181 | 7 |
| 2 | Aarav Patel  | 23SS02IT101 | 8 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
  
```